NEWTON SCHOOL

PRE-CALCULUS | TEST PREP

Induction Practice Set — PCBCH Problems 6 to 12

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Standard: *Mathematical Induction (regular + strong) · AP Calculus BC / Pre-Calc — Sequences, Recursive Definitions, Number Theory, Combinatorial Identities.*

How to read this set

Every problem is a Mr. Jones *induction* exercise. Each one shows three steps in the green Solution box: **Base case**, **Inductive Hypothesis (IH)**, and **Inductive Step**. Master the *skeleton* first — the algebra after that is just bookkeeping.

Problem 6 Infinite Nested Radical Is Always Less Than 2

STATEMENT

Prove that every member of the sequence $\sqrt{2},\ \sqrt{2+\sqrt{2}},\ \sqrt{2+\sqrt{2+\sqrt{2}}},\ \ldots$ is less than 2.

SOLUTION

Setup. Define recursively: $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$. We prove $a_n < 2$ for all $n \geq 1$ by induction.

Base ($n = 1$). $a_1 = \sqrt{2} \approx 1.414 < 2$. ✓

IH. Assume $a_k < 2$.

Step. Then $a_{k+1} = \sqrt{2 + a_k} < \sqrt{2+2} = \sqrt{4} = 2$. ✓
(Square root is increasing, so $a_k < 2 \Rightarrow 2 + a_k < 4 \Rightarrow \sqrt{2 + a_k} < 2$.)

Conclusion. By induction, $a_n < 2$ for every $n \geq 1$. ■

Problem 7 Tribonacci-like Sequence Grows Slower Than 2^n

STATEMENT

Consider $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ with $a_1 = 1,\ a_2 = 2,\ a_3 = 3$. Prove that for all positive integers n, $a_n < 2^n$.

SOLUTION (STRONG INDUCTION)

Base ($n = 1, 2, 3$). $a_1 = 1 < 2 = 2^1$, $a_2 = 2 < 4 = 2^2$, $a_3 = 3 < 8 = 2^3$. ✓
(We need three base cases because the recurrence reaches back three steps.)

Strong IH. Assume $a_j < 2^j$ for every $1 \le j \le k$, where $k \ge 3$.

Step. Using the recurrence and IH on the three previous terms:

$$a_{k+1} \;=\; a_k + a_{k-1} + a_{k-2} \;<\; 2^k + 2^{k-1} + 2^{k-2}.$$

Factor 2^{k-2}: $2^k + 2^{k-1} + 2^{k-2} = 2^{k-2}(4 + 2 + 1) = 7 \cdot 2^{k-2}$.
Since $7 < 8$: $7 \cdot 2^{k-2} < 8 \cdot 2^{k-2} = 2^{k+1}$. ✓

Conclusion. $a_n < 2^n$ for all $n \ge 1$. ■

Insight. Each Tribonacci term is the sum of the previous three, and the bound 2^n doubles at every step. The factor-7 vs 8 gap is the slack that lets the inequality stay strict.

Problem 8 Chicken McNugget — Postage ≥ 8 Cents

STATEMENT

Prove that using only 3-cent and 5-cent stamps, one can generate any postage ≥ 8 cents.

SOLUTION (STRONG INDUCTION)

Base cases ($n = 8, 9, 10$). $8 = 3 + 5, \quad 9 = 3 + 3 + 3, \quad 10 = 5 + 5$. ✓
(We need three consecutive base cases because the inductive step jumps back by 3.)

Strong IH. Every integer j with $8 \le j \le k$ can be formed (for $k \ge 10$).

Step. For $k + 1$, write $k + 1 = (k - 2) + 3$.
Since $k \ge 10$, we have $k - 2 \ge 8$, so by the strong IH the value $(k - 2)$ cents can be made from 3¢ and 5¢ stamps. Add one 3¢ stamp to get $k + 1$. ✓

Conclusion. Every postage ≥ 8 cents can be made. ■

Note. The integers that **cannot** be expressed are exactly $\{1, 2, 4, 7\}$. The largest one (7) is the *Frobenius number* for the coin set $\{3, 5\}$: $g(3, 5) = 3 \cdot 5 - 3 - 5 = 7$.

Problem 9 L-Tromino Tiling of a $2^n \times 2^n$ Board (One Corner Removed)

STATEMENT

For every positive integer n, prove that a $2^n \times 2^n$ checkerboard with **one** corner square removed can be tiled by rotatable L-shaped 3-square pieces (L-trominoes).

SOLUTION (INDUCTION ON n)

Base ($n = 1$). A 2×2 board minus one corner has exactly 3 squares — that is precisely one L-tromino. ✓

IH. Every $2^k \times 2^k$ board missing one corner square can be tiled with L-trominoes.

Step ($n = k + 1$). Take a $2^{k+1} \times 2^{k+1}$ board with one corner missing. Split it into **four** $2^k \times 2^k$ **quadrants**.

- The missing corner lies in **exactly one** quadrant. Apply the IH directly to that quadrant.
- For the other three quadrants: place **one** L-tromino at the very center of the big board so it covers the inner corner cell of each of those three quadrants.
- Now each of those three quadrants effectively has a "missing" cell at its inner corner — so the IH applies to each. ✓

Conclusion. By induction, the tiling exists for every $n \geq 1$. ■

Picture in your head. The central L-tromino is the *"seed"* — it converts the problem of "three full quadrants + one missing-corner quadrant" into "four missing-corner quadrants", and then the IH closes the deal.

Fibonacci Preface — for Problems 10, 11, 12

The Fibonacci sequence is defined by $F_n = F_{n-1} + F_{n-2}$ with $F_1 = 1,\ F_2 = 1$. First values: $F_3 = 2,\ F_4 = 3,\ F_5 = 5,\ F_6 = 8,\ F_7 = 13,\ F_8 = 21,\ \ldots$

Problem 10 $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$

STATEMENT

Prove that for all positive integers n, $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$.

SOLUTION

Base ($n = 1$). LHS $= F_1 = 1$. RHS $= F_3 - 1 = 2 - 1 = 1$. ✓

IH. $F_1 + F_2 + \cdots + F_k = F_{k+2} - 1$.

Step. Add F_{k+1} to both sides of IH:

$$F_1 + \cdots + F_k + F_{k+1} \;=\; (F_{k+2} - 1) + F_{k+1} \;=\; (F_{k+1} + F_{k+2}) - 1 \;=\; F_{k+3} - 1.$$

We used the Fibonacci recurrence $F_{k+3} = F_{k+1} + F_{k+2}$. ✓

Conclusion. $\displaystyle\sum_{i=1}^{n} F_i = F_{n+2} - 1$ for all $n \geq 1$. ■

Problem 11 Cassini's Identity — $F_{n-1}F_{n+1} - F_n^2 = (-1)^n$

STATEMENT

Prove that for all positive integers $n \geq 2$, $F_{n-1} \cdot F_{n+1} - (F_n)^2 = (-1)^n$.

SOLUTION

Base ($n = 2$). $F_1F_3 - F_2^2 = 1 \cdot 2 - 1^2 = 1 = (-1)^2$. ✓

IH. $F_{k-1}F_{k+1} - F_k^2 = (-1)^k$.

Step. Compute $F_kF_{k+2} - F_{k+1}^2$:

- $F_{k+2} = F_{k+1} + F_k \;\Rightarrow\; F_kF_{k+2} = F_kF_{k+1} + F_k^2$.
- $F_{k+1}^2 = F_{k+1}(F_k + F_{k-1}) = F_kF_{k+1} + F_{k-1}F_{k+1}$.
- Subtract:
  $F_kF_{k+2} - F_{k+1}^2 = F_k^2 - F_{k-1}F_{k+1} = -(F_{k-1}F_{k+1} - F_k^2) = -(-1)^k = (-1)^{k+1}$.
  ✓

Conclusion. Cassini's identity holds for every $n \geq 2$. ■

Why the alternating sign? Multiplying by -1 at every inductive step flips the parity — that is exactly $(-1)^{k+1} = -(-1)^k$.

Problem 12 $\sum_{k=1}^{2n-1} F_kF_{k+1} = F_{2n}^2$

STATEMENT

Prove that for all positive integers n, $F_1F_2 + F_2F_3 + F_3F_4 + \cdots + F_{2n-1}F_{2n} = (F_{2n})^2$.

SOLUTION

Base ($n = 1$). LHS $= F_1F_2 = 1 \cdot 1 = 1$. RHS $= F_2^2 = 1$. ✓

IH. $\displaystyle\sum_{k=1}^{2n-1} F_kF_{k+1} = F_{2n}^2$.

Step (advance from n to $n+1$ = add the next *two* terms).

$$\sum_{k=1}^{2(n+1)-1} F_kF_{k+1} \;=\; F_{2n}^2 + F_{2n}F_{2n+1} + F_{2n+1}F_{2n+2}.$$

Group the first two terms by factoring F_{2n}:

$$F_{2n}^2 + F_{2n}F_{2n+1} \;=\; F_{2n}(F_{2n} + F_{2n+1}) \;=\; F_{2n}F_{2n+2}.$$

Now factor F_{2n+2} from what remains:

$$F_{2n}F_{2n+2} + F_{2n+1}F_{2n+2} \;=\; F_{2n+2}(F_{2n} + F_{2n+1}) \;=\; F_{2n+2} \cdot F_{2n+2} \;=\; F_{2n+2}^2.$$

✓ which is $F_{2(n+1)}^2$.

Conclusion. The identity holds for every $n \geq 1$. ■

Two-term jump. Because the index on the LHS is $2n-1$, advancing from n to $n+1$ adds **two** new summands ($F_{2n}F_{2n+1}$ and $F_{2n+1}F_{2n+2}$), not one. The Fibonacci recurrence is then applied **twice** in the algebra.

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